

Exercises (with Solutions) · Chapter 4

1. Problem

Consider the nonparametric regression model

$$Y_i = f(X_i) + \epsilon_i, \quad i = 1, \dots, n,$$

where $X_i = i/n$, the error terms ϵ_i are i.i.d. with $E(\epsilon_i) = 0$ and $Var(\epsilon_i) = \sigma^2 > 0$, and f is an unknown smooth function. Suppose we estimate f using the KNN-regression estimator

$$\hat{f}_K(x_0) = \frac{1}{K} \sum_{i \in \mathcal{N}_0} Y_i.$$

at a fixed point $x_0 \in (0, 1)$, where $\mathcal{N}_0$ is the set of indices corresponding to the K nearest neighbors of x_0 among the design points $X_1, \dots, X_n$.

(a) Compute the bias $\text{Bias}(\hat{f}_K(x_0)) = E(\hat{f}_K(x_0)) - f(x_0)$ of the KNN-regression estimator $\hat{f}_K(x_0)$.

(b) Compute the variance $Var(\hat{f}_K(x_0))$ of the KNN-regression estimator $\hat{f}_K(x_0)$.

(c) Now, let $f(x) = 5 + 3 \sin(2\pi x)$, $\sigma^2 = 1$, and $n = 100$.

Compute the bias of the KNN-regression estimator $\hat{f}_K(x_0)$ at $x_0 = 0.25$ for $K = 5$ and $K = 20$.

(d) Consider the same setup as in the previous question.

Compute the variance of the KNN-regression estimator $\hat{f}_K(x_0)$ at $x_0 = 0.25$ for $K = 5$ and $K = 20$.

(e) Consider the same setup as in the previous question.

Compute the local mean squared estimation error of the KNN-regression estimator $\hat{f}_K(x_0)$ at $x_0 = 0.25$ for $K = 5$ and $K = 20$.

(f) Consider the same setup as in the previous question.

Compute the local mean squared prediction error of the KNN-regression estimator $\hat{f}_K(x_0)$ at $x_0 = 0.25$ for $K = 5$ and $K = 20$.

Solution

(a) The bias of the KNN-regression estimator $\hat{f}_K(x_0)$ is given by

$$\text{Bias}(\hat{f}_K(x_0)) = E(\hat{f}_K(x_0)) - f(x_0)$$

So, we mainly have to compute the mean $E(\hat{f}_K(x_0))$:

$$\begin{aligned}
 E(\hat{f}_K(x_0)) &= E\left(\frac{1}{K} \sum_{i \in \mathcal{N}_0} Y_i\right) \\
 &= \frac{1}{K} \sum_{i \in \mathcal{N}_0} E(Y_i) \\
 &= \frac{1}{K} \sum_{i \in \mathcal{N}_0} E(f(X_i) + \epsilon_i) \\
 &= \frac{1}{K} \sum_{i \in \mathcal{N}_0} E(f(X_i)) + \frac{1}{K} \sum_{i \in \mathcal{N}_0} E(\epsilon_i) \\
 &= \frac{1}{K} \sum_{i \in \mathcal{N}_0} E(f(X_i)) + \frac{1}{K} \sum_{i \in \mathcal{N}_0} 0 \\
 &= \frac{1}{K} \sum_{i \in \mathcal{N}_0} E(f(i/n)) \\
 &= \frac{1}{K} \sum_{i \in \mathcal{N}_0} f(i/n)
 \end{aligned}$$

Thus, the bias is

$$\text{Bias}(\hat{f}_K(x_0)) = \frac{1}{K} \sum_{i \in \mathcal{N}_0} f(i/n) - f(x_0)$$

(b) The variance of the KNN-regression estimator $\hat{f}_K(x_0)$ is given by

$$\begin{aligned}
 \text{Var}(\hat{f}_K(x_0)) &= \text{Var}\left(\frac{1}{K} \sum_{i \in \mathcal{N}_0} Y_i\right) \\
 &= \frac{1}{K^2} \text{Var}\left(\sum_{i \in \mathcal{N}_0} Y_i\right) \\
 &= \frac{1}{K^2} \sum_{i \in \mathcal{N}_0} \text{Var}(Y_i) \\
 &= \frac{1}{K^2} \sum_{i \in \mathcal{N}_0} \text{Var}(f(X_i) + \epsilon_i) \\
 &= \frac{1}{K^2} \sum_{i \in \mathcal{N}_0} \text{Var}(f(i/n) + \epsilon_i) \\
 &= \frac{1}{K^2} \sum_{i \in \mathcal{N}_0} \text{Var}(\epsilon_i) \\
 &= \frac{1}{K^2} \sum_{i \in \mathcal{N}_0} \sigma^2 \\
 &= \frac{K}{K^2} \sigma^2 \\
 &= \frac{1}{K} \sigma^2
 \end{aligned}$$

(c) Given $f(X_i) = 5 + 3 \sin(2\pi X_i)$, $x_0 = 0.25$, and $n = 100$, we first identify the K nearest neighbors of $x_0 = 0.25$. The design points are $X_i = i/100$ for $i = 1, \dots, 100$.

- The nearest neighbors of $x_0 = 0.25$ for $K = 5$ are:

$$X_{23} = 0.23, X_{24} = 0.24, X_{25} = 0.25, X_{26} = 0.26, X_{27} = 0.27.$$

Thus the bias for $K = 5$ is

$$\begin{aligned} \text{Bias}(\hat{f}_5(0.25)) &= \frac{1}{5} (f(0.23) + f(0.24) + f(0.25) + f(0.26) + f(0.27)) - f(0.25) \\ &= 7.988 - 8 = -0.012 \end{aligned}$$

- The nearest neighbors of $x_0 = 0.25$ for $K = 20$ are:

$$X_{16} = 0.16, X_{17} = 0.17, X_{18} = 0.18, \dots, X_{35} = 0.35.$$

Thus the bias for $K = 20$ is

$$\begin{aligned} \text{Bias}(\hat{f}_{20}(0.25)) &= \frac{1}{20} (f(0.16) + f(0.17) + f(0.18) + \dots + f(0.35)) - f(0.25) \\ &= 7.806 - 8 = -0.194 \end{aligned}$$

- (d) Given $\sigma^2 = 1$, the variance of the KNN-regression estimator $\hat{f}_K(x_0)$ for $K = 5$ and $K = 20$ are computed as follows:

- For $K = 5$: $\text{Var}(\hat{f}_K(x_0)) = \frac{1}{K} \sigma^2 = \frac{1}{5}$

- For $K = 20$: $\text{Var}(\hat{f}_K(x_0)) = \frac{1}{K} \sigma^2 = \frac{1}{20}$

- (e) The local mean squared estimation error is given by

$$\text{Bias}(\hat{f}_K(x_0))^2 + \text{Var}(\hat{f}_K(x_0)).$$

Thus, for $x_0 = 0.25$ we have:

- For $K = 5$: $0.00014 + \frac{1}{5} = 0.20014$

- For $K = 20$: $0.03764 + \frac{1}{20} = 0.08764$

- (f) The local mean squared prediction error is given by

$$\text{Bias}(\hat{f}_K(x_0))^2 + \text{Var}(\hat{f}_K(x_0)) + \sigma^2.$$

Thus, for $x_0 = 0.25$ we have:

- For $K = 5$: $0.00014 + \frac{1}{5} + 1 = 1.20014$

- For $K = 20$: $0.03764 + \frac{1}{20} + 1 = 1.08764$

2. Problem

Curse of Dimensionality: The accuracy of nonparametric methods typically deteriorates as the dimension $p \geq 1$ of the predictor variables $X_i = (X_{i1}, \dots, X_{ip})^\top$ increases. This phenomenon is known as the "curse of dimensionality". The reason for this problem is the "emptiness" of high-dimensional spaces. To illustrate this, consider the case of a p -dimensional predictor variable $X_i = (X_{i1}, \dots, X_{ip})^\top$, with standard normal components $X_{ij} \sim \mathcal{N}(0, 1)$ for all $j = 1, \dots, p$ all independent from each other; i.e. X_{ij} and X_{ik} are independent for all $j \neq k$.

- (a) Consider the case of $p = 1$: Compute the probability that X_i lies in the interval $[-1, 1]$.
 (b) Consider the case of $p = 2$: Compute the probability that X_i lies in the square $[-1, 1]^2$.
 (c) Consider the general case of $p \geq 1$: Compute the probability that X_i lies in $[-1, 1]^p$.
 (d) Consider a sample size of $n = 100$. How many data points do you expect to see in $[-1, 1]^p$ for $p \in \{1, 5, 15\}$?

Solution

- (a) For $p = 1$, we have $X_i = X_{i1} \sim \mathcal{N}(0, 1)$. Thus, we need to compute

$$\begin{aligned}\mathbb{P}(-1 \leq X_{i1} \leq 1) &= \Phi(1) - \Phi(-1) \\ &= 2\Phi(1) - 1 \\ &\approx 2 \cdot 0.841 - 1 = 0.682,\end{aligned}$$

where Φ denotes the cumulative distribution function of the standard normal distribution.

- (b) For $p = 2$, we have $X_i = (X_{i1}, X_{i2})^\top$ with $X_{i1} \sim \mathcal{N}(0, 1)$ and $X_{i2} \sim \mathcal{N}(0, 1)$, where X_{i1} and X_{i2} are independent of each other. Thus, we need to compute

$$\begin{aligned}\mathbb{P}(-1 \leq X_{i1} \leq 1 \text{ and } -1 \leq X_{i2} \leq 1) &= \mathbb{P}(-1 \leq X_{i1} \leq 1) \cdot \mathbb{P}(-1 \leq X_{i2} \leq 1) \\ &= (2\Phi(1) - 1)^2 \\ &\approx (2 \cdot 0.841 - 1)^2 = 0.465,\end{aligned}$$

- (c) For $p \geq 1$, we have that

$$\begin{aligned}\mathbb{P}(-1 \leq X_{i1} \leq 1 \text{ and } -1 \leq X_{i2} \leq 1 \text{ and } \dots \text{ and } -1 \leq X_{ip} \leq 1) \\ &= \mathbb{P}(-1 \leq X_{i1} \leq 1) \cdot \mathbb{P}(-1 \leq X_{i2} \leq 1) \cdot \dots \cdot \mathbb{P}(-1 \leq X_{ip} \leq 1) \\ &= (2\Phi(1) - 1)^p \\ &\approx (2 \cdot 0.841 - 1)^p\end{aligned}$$

- (d) The expected number of data points in $[-1, 1]^p$ is given by

$$n \cdot (2\Phi(1) - 1)^p \approx 100(2 \cdot 0.841 - 1)^p.$$

Thus

- for $p = 1$, we expect to see approximately $100 \cdot (2 \cdot 0.841 - 1)^1 = 68.2$ data points,
- for $p = 5$, we expect to see approximately $100 \cdot (2 \cdot 0.841 - 1)^5 = 14.6$ data points, and
- for $p = 15$, we expect to see approximately $100 \cdot (2 \cdot 0.841 - 1)^{15} = 0.3$ data points.